Dambi Dollo University

Research, Publication, and Ethics Directorate Office

Concept Note / Synopsis

THEME 2: Science, Technology, and Innovation

Sub-Theme: AI and Machine Learning

Title: Deep Learning-Based Identification of Unsafe Housing Structures in Dambi Dollo Using High-Resolution Satellite Imagery

Authors:

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Introduction

Housing safety is a critical aspect of urban resilience, public health, and disaster preparedness. In rapidly growing Ethiopian towns such as Dambi Dollo, informal settlements and substandard housing have become increasingly common due to urban expansion, limited regulation, and insufficient monitoring. Unsafe structures including poorly constructed houses, deteriorating roofs, makeshift shelters, and buildings at risk of collapse pose significant risks to residents, especially during extreme weather events or natural hazards.

Traditional assessment methods that rely on field surveys are labor-intensive, slow, and lack the coverage needed for continuous monitoring. Satellite imagery, combined with

advances in artificial intelligence particularly deep learning offers a powerful solution for identifying unsafe housing structures at scale. However, in Ethiopia, the use of AI-driven remote sensing for housing safety assessment remains limited. This research proposes a deep learning-based system to analyze high-resolution satellite imagery and detect unsafe housing structures across Dambi Dollo city. The outcome will support urban planners, disaster risk managers, and local authorities in making evidence-based decisions.

General Objective

To develop a deep learning-based system for identifying and mapping unsafe housing structures in Dambi Dollo city using high-resolution satellite imagery.

Specific Objectives

- To collect and preprocess high-resolution satellite data covering Dambi Dollo city for housing structure analysis.
- To annotate and classify housing structures into safe and unsafe categories based on visual characteristics.
- To train and validate a deep learning model capable of detecting unsafe structures from imagery.
- To generate a spatial map of unsafe housing structures for urban planning and emergency preparedness.
- To evaluate the performance and reliability of the developed detection system.

Statement of the Problem

Dambi Dollo faces growing challenges related to housing safety, driven by rapid urbanization, informal construction practices, and limited monitoring capacity. Many residential structures are built using low-quality materials or without proper engineering standards, increasing their vulnerability to collapse, fire hazards, and extreme weather events. Municipal authorities lack a reliable, up-to-date, and spatially comprehensive system for assessing housing risk. Current methods depend heavily on manual inspection, which is time-consuming and often fails to capture the dynamic nature of urban growth.

Although deep learning and remote sensing technologies have demonstrated strong potential in infrastructure assessment globally, their application in Ethiopian cities

remains minimal. There is a critical research gap in using AI-enhanced satellite imagery to automate the detection of unsafe housing. Addressing this gap will enable local authorities to prioritize interventions, improve disaster preparedness, and enhance community safety.

Expected Products / Outcomes

- A high-resolution map of unsafe housing structures across Dambi Dollo city.
- A trained deep learning model for detecting unsafe housing from satellite imagery.
- A GIS-integrated dataset for urban planning and emergency response.
- A guideline for applying AI and remote sensing in housing safety assessment.

Beneficiaries / Customers

- Municipal authorities and urban planners.
- Emergency and disaster response agencies.
- Local communities vulnerable to unsafe housing risks.
- NGOs working in urban development and resilience.

Brief Method / Layout

The study integrates high-resolution satellite imagery with deep learning to identify unsafe housing structures. Imagery will be collected from Google Earth Pro, Mapbox, or other high-resolution sources and preprocessed for clarity. Housing structures will be manually annotated and categorized based on safety indicators. A convolutional neural network (CNN) model (e.g., YOLOv8 or UNet) will be trained to detect unsafe housing characteristics such as deteriorated roofs, visible structural damage, and informal construction. The detected structures will be exported into GIS software (QGIS/ArcGIS) to create a detailed risk map. The system will be evaluated for accuracy and reliability, and recommendations will be provided for city-level adoption.

Flow Chart

1. **Data Acquisition**: Collect satellite images and optional drone photos.
2. **Preprocessing & Annotation**: Clip images, enhance quality, and annotate houses.
3. **Model Development**: Train YOLOv8/UNet for unsafe-house detection.
4. **Validation**: Assess accuracy, precision, recall, and IoU.
5. **Mapping**: Export detections into GIS to create an unsafe housing map.
6. **Recommendation**: Provide actionable safety insights for authorities.

Collaboration with Industries and Institutes

This research will be conducted in collaboration with Dambi Dollo municipal authorities, local community leaders, and the Dambi Dollo University Department of Information Technology to support data collection, validation, and practical implementation.

Estimated Budget

Estimated total cost: 380,000 ETB

Expected Number of Publications

2